

WALIYA SHAH

Data Engineer | Data Analyst | AI & Automation

(945) 527-4693 | waliyashah00@gmail.com | [linkedin.com/in/waliyashah](https://www.linkedin.com/in/waliyashah) | github.com/waliyashah

Springfield, MA | Open to Relocation & Remote

PROFESSIONAL SUMMARY

Data Engineer and Analyst with 5+ years of experience building ETL pipelines, SQL data platforms, and BI dashboards across retail, energy, and healthcare. Skilled in Python, SQL, AWS, Tableau, and Power BI, with hands-on NLP and LLM experience. Processed 2M+ monthly transactions, cut data errors by 80%, and automated reporting workflows saving 50+ hours/month for business stakeholders.

TECHNICAL SKILLS

AI / ML: Hugging Face Transformers, PyTorch, TensorFlow/Keras, LLM Integration, RAG, NLP, Deep Learning (CNNs), Scikit-learn, Multi-Agent Systems

Languages & Engineering: Python (Pandas, NumPy, Flask, Streamlit), SQL, R, Apache Airflow, dbt, Spark (PySpark), Kafka, Docker, ETL/ELT, Pipeline Orchestration, Web Scraping, REST APIs, CI/CD (GitHub Actions)

Cloud & Databases: AWS (S3, Redshift, Glue, Lambda, Athena), Snowflake, Databricks, Terraform, SQL Server, PostgreSQL, MySQL, MongoDB, BigQuery, Data Warehousing, Data Modeling, Query Optimization

BI & Analytics: Tableau, Power BI, Looker, Excel (DAX, Power Query), Statistical Modeling, A/B Testing, Regression, Forecasting, Git, JIRA, Agile/Scrum

PROFESSIONAL EXPERIENCE

Data & Automation Engineer (Research Assistant)

Aug 2025 – Present

Western New England University, Springfield, MA

- Engineered automated credential-management workflows with Hyperledger Aries and Python, securing issuance and verification of 1,000+ academic and healthcare credentials across distributed identity networks.
- Orchestrated Dockerized multi-agent workflows with REST API coordination between autonomous agents, improving processing reliability by 40% and reducing manual steps by 60%.
- Built a full-stack self-sovereign identity (SSI) application with verification-gated access, where patients and doctors see only their own records, cutting credential validation time by 50%.

Data Analyst Intern

Jan 2025 – May 2025

Westfield Gas & Electric, Westfield, MA

- Designed a centralized SQL Server data platform replacing a document-driven regulatory tracking system, supporting 10+ departments and self-service analytics for 99+ users.
- Developed Python ETL pipelines applying NLP-based text extraction to transform 100+ unstructured Word documents into analytics-ready relational tables.
- Built a full-stack Flask web app with role-based access control, approval workflows, and audit logging; automated KPI reporting saving 30+ hours/month at 99% accuracy.

Data Engineer (Research Assistant) — NLP & AI Pipelines

Mar 2024 – Aug 2024

Western New England University, Springfield, MA

- Built Python web-scraping pipelines consolidating 20,000+ FDA drug records monthly from fragmented sources, improving structured data accessibility by 85%.
- Cleaned and standardized text-heavy FDA datasets into relational schemas powering an NLP-driven multilingual AI chatbot for natural-language drug queries.
- Scraped and unified 10,000+ IEEE Xplore and arXiv papers into a searchable knowledge base feeding NLP research pipelines, cutting literature review effort by 60%.

Senior Data Analyst

Jan 2020 – May 2023

SnappRetail (Retail Tech Startup), Pakistan

- Designed and maintained AWS ETL pipelines (S3, Redshift, Glue) processing 2M+ monthly POS transactions, reducing data discrepancies by 80% and improving demand-forecast reliability.
- Built 20+ executive Tableau dashboards used daily by 50+ stakeholders; engineered features for demand-forecasting models.
- Automated 30+ recurring reports with Python (saving 50+ hours/month) and built competitor-price web scrapers tracking 500+ SKUs weekly to drive pricing decisions.

AI PROJECTS & PUBLICATIONS

- Multilingual AI Chatbot (Hugging Face, PyTorch):** Built a conversational assistant combining a GPT-2-based LLM with mBART-50 multilingual translation, enabling cross-language question answering with GPU-accelerated inference.
- SEC EDGAR Filings Data Pipeline (Python, Pandas):** Automated download and parsing of SEC EDGAR index files into searchable datasets queryable by company, CIK, and filing period, with time-series analysis and visualization of filing trends.
- HR Analytics Dashboard:** Interactive dashboard visualizing 22K+ employee records across demographics, hiring trends, and geography.
- Research Papers:** (1) CNN-based plant disease detection from drone imagery (deep learning, Convolutional Autoencoder); (2) AI in Finance — AI-driven trading, market stability, and regulatory risk.

EDUCATION & CERTIFICATIONS

M.S. Computer Science, GPA 3.8 — Western New England University, Springfield, MA

May 2025

Certifications: [Google Data Analytics \(Coursera\)](https://www.coursera.org/certificate/google-data-analytics) | [BI with BigQuery \(Google Cloud\)](https://www.google.com/cloud/certifications/bigquery/) | [Certified Data Analyst \(FAST-NU\)](https://www.fast-nu.com/certified-data-analyst/)